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Physical activity and dietary habits related to cardiovascular risk in independent community-living older women

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ABSTRACT

A side effect of increased life expectancy is a surge in sequelae of diseases and injuries, which in turn increase the duration of life with disability among the elderly. The aim of this study was to ascertain the physical activity and nutritional parameters that better predict cardiovascular risk in a cohort of older women. A cross-sectional study was designed including 65 women able to independently perform basic activities of daily life. Data collection included anthropometric measurement, blood pressure measurement, blood analytics, objectively measurement of physical activity, and dietary assessment. We were able to generate models that explain the relationship between physical activity, diet, and these health measurement parameters. We observed that the combination of moderate physical activity and a diet including protein-rich foods as nuts, dairy, and eggs were better predictors associated with favorable changes in risk factors for cardiovascular disease than age.

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1. Introduction

A side effect of increased life expectancy is a surge in sequelae of diseases and injuries, which in turn increase the duration of life with disability among the elderly. Non-fatal dimensions of disease and injury require attention from health systems and are a major expense as years lived with disability rise while mortality declines (Vos et al., 2015).

Some of the leading causes of stunted lifespan are cardiovascular and circulatory diseases (Lozano et al., 2013). The impact of lifestyle on the risk of cardiovascular disease (CVD) has been well established in numerous studies (Mozaffarian et al., 2008). Physical exercise and diet are highly effective in facilitating coronary risk reduction, complementing and enhancing medications, and in some instances even outperforming drug therapy (Franklin et al., 2014). Moreover, physical inactivity and dietary risk factors collectively accounted for 10% of global disability adjusted life years (DALYs) in 2010 (Lim et al., 2013). To tackle this situation, health policy should be addressed to empower people to develop healthy life habits that will reduce CVD-related morbidity and disability.

There is strong scientific evidence in support of the associations between diet, physical activity, and cardiovascular risk. However, both diet and physical activity are rarely analyzed together. In addition, in most of the previous studies, physical activity was measured using questionnaires while our study utilized accelerometer. Accelerometers provide a means of objectively quantifying total daily activity, they are particularly sensitive for detecting time spent on various intensities of physical activity, and they are acceptable to seniors (Harris et al., 2009). One of the strengths of our study is that to our knowledge, this is the first study to analyze the effects of physical activity, measured using accelerometers, and diet together on variables related to cardiovascular risk in older adults.

The aim of our study was to ascertain the physical activity and nutritional parameters that better predict cardiovascular risk in a cohort of older women. Our results will assist in the design and implementation of more effective interventions to promote healthy aging.

2. Methods

2.1. Sampling and recruitment

The study included 65 women who were able to perform basic activities of daily life independently and lived in their own home. The

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inclusion criteria were: old adults, aged ≥ 60 years who were all capable of performing basic daily activities by themselves and lived on their own. The exclusion criteria were: individuals with cognitive impairment, with a dementia diagnosis or mentally disabled. They were recruited in our collaborating senior centers situated in San Sebastian ("Gure Egunsentia" senior center and "Aulas de la Experiencia" college for adults) and Zarautz ("Udaberri" senior center) in Spain. These centers were chosen on the basis of the willingness to collaborate of their members. All participants gave written informed consent, and the study was approved by the Ethics Committee of the "Matia Instituto Gerontológico" Foundation.

2.2. Anthropometric measurement

Weight (kg), height (cm), and waist circumference were measured using height boards and weighing scales (Añó Sayol SL, Barcelona, Spain). The weight and height of the participants were then used to calculate body mass index (BMI) in kg/m^2. All measurements were taken by the same person following the standards of the International Society for the Advancement of Kinanthropometry (Marfell-Jones et al., 2012).

2.3. Blood pressure measurement

Blood pressure was measured in the right arm with an automatic sphygmomanometer (Omron M5-1, OMRON Corporation, Japan) while the participant was in a seated position. All measurements were taken by the same person following a validated protocol (Panamerican Hypertension Initiative, 2003).

2.4. Analytical procedures

Blood samples were obtained after a minimum of eight hours fasting, with the subject in a sitting position. Measurement of fasting plasma glucose, total cholesterol, high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol, and triglycerides were performed using standard clinical laboratory methods. Serum parameters were measured with COBAS 711 and 6000 analyzers (Roche Diagnostics, Mannheim, Germany).

2.5. Measurement of physical activity

To accurately measure physical activity we used accelerometers that record active and sedentary periods during everyday life. Accelerometers offer a number of desirable features for monitoring human movement: amount, duration, frequency, and intensity of movement (Mathie et al., 2004). In this study, accelerometry was performed using the Actigraph GT3X model (Actigraph LLC, Pensacola, FL, USA). The monitor is worn on the hip with a belt and participants were asked to wear one for a five-day period removing it only for sleeping and bathing. All the participants wore the accelerometers from Tuesday to Saturday to take into account also, at least, the activity done during one of the days of the weekend.

Activity was recorded using 10-second epochs. Data files recorded on the accelerometers were downloaded and processed with Actilife software (version 5, Actigraph, 2011). For the analysis presented here, only days on which the monitors were worn for 10 or more hours were considered valid and it was required that there were at least three days of data to validate the recorded data (Hart et al., 2011). Non-wear time was defined by an interval of at least 60 consecutive minutes of zero activity intensity counts, with tolerance for 2 min of counts between 0 and 50. The summary variables presented in this analysis were as follows: number of steps per day, mean counts per minute (cpm), and number of minutes per day spent in intensity-specific categories. It is difficult to determine cut-off points to classify the intensity of physical activity in older people because there is currently no consensus in the scientific literature. In the present study the cut-off points to classify the intensity of physical activity follow the classification developed by Freedson et al. (1998), where the interval for moderate activity was set between 1952 and 5724 cpm and the vigorous activity was defined as all activity ≥ 5725 cpm; so moderate to vigorous physical activity (MVPA) was defined as all activity ≥ 1952 cpm (Freedson et al., 1998). To classify lower intensity activities, we took as reference the article by Lohne Seiler et al. (2014). Thus, the following intensity-specific cut-off points were applied to the raw data: sedentary time was defined as all activity below 100 cpm, low-intensity physical activity was defined as all activity between 100 and 759 cpm, and time in lifestyle activity was defined as all activity between 760 and 1951 cpm (Lohne Seiler et al., 2014). The number of minutes per day at different intensities was determined by adding all minutes where the count met the criterion for the specific intensity divided by the number of valid days.

2.6. Dietary assessment

The short food-frequency quantitative (FFQ) questionnaire was used to assess dietary intake. The central question was, "How often did you eat or drink the following food items?" Forty-five items were included in the questionnaire and were grouped into several food groups (vegetables, fruits, meat, etc.). The FFQ questionnaire has been previously utilized successfully in Spain (Trinidad Rodríguez et al., 2008).

2.7. Statistical analysis

All statistical analyses were performed using the Statistical Package for the Social Sciences (SPSS Inc., version 21). The Kolmogorov-Smirnov test was used to check the normality of the data. Correlations among variables were analyzed using Pearson's and Spearman's coefficients (for non-parametric distributions). Afterwards, variables that were statistically significant were introduced into linear regression models. The variables that did not present a normal distribution were also categorized and probed in both ways inside the models.

Significant linear regression models were searched to find a statistical association with parameters related to cardiovascular risk. Backward selection was used to select the model that best explained the variability of the dependent variables. First, standardized coefficients were studied to measure the contribution of each independent variable. Then, the presence of collinearity was intensively studied to explain the relationship among variables using Pearson's correlation test, Variance Inflation Factor (VIF), and condition index. In order to test the models, a Durbin-Watson test was used to discard a hypothetical nonparametric distribution of residuals. The best explanatory models were included in this study.

The glucose variable followed a non-parametric distribution, so a logistic regression was used to test the association among variables. Differences were considered to be statistically significant when $p < 0.05$.

3. Results

Sixty-five women aged ≥ 60 years with mean age of 70.1 and a standard deviation of 6.6 years were enrolled in the study. Anthropometric characteristics, blood pressure, analytical parameters, and variables of physical activity are shown in Table 1. Dietary data are shown in Table 2.

3.1. Correlations

Counts per minute (cpm), steps per day, and moderate intensity physical activity were negatively correlated with BMI, waist circumference, and fasting plasma glucose concentrations and positively correlated with blood HDL-C levels (Table 3). Diastolic blood pressure showed a negative correlation with time spent in moderate physical activity, but not with cpm or the number of steps per day. The practice of lower

Table 1
Anthropometric characteristics, blood pressure, analytical parameters, and physical activity data for a cohort of older women (n = 65).

Variable[a]	Mean ± SD
Anthropometric characteristics	
Age (years)	70.1 ± 6.60
Weight (kg)	65.8 ± 10.4
Height (cm)	155 ± 5.90
BMI (kg/m^2)	27.3 ± 3.72
WC (cm)	87.7 ± 9.14
Blood pressure	
SBP (mm Hg)	145 ± 18.5
DBP (mm Hg)	77.5 ± 9.26
Analytical parameters	
Glucose (mg/dL)	90.1 ± 15.5
Total cholesterol (mg/dL)	229 ± 28.6
HDL-C (mg/dL)	67.0 ± 16.0
LDL-C (mg/dL)	144 ± 25.4
Triglycerides (mg/dL)	86.6 ± 39.9
Variables of physical activity[b]	
Counts per minute (cpm)	262 ± 117
Steps per day	11,850 ± 4574
Sedentary PA (min/day)	521 ± 65.5
Low-intensity PA (min/day)	136 ± 36.6
Lifestyle PA (min/day)	70.2 ± 21.0
Moderate PA (min/day)	71.2 ± 37.5
Vigorous PA (min/day)	1.64 ± 5.45
MVPA (min/day)	72.8 ± 39.2

[a] BMI, body mass index; WC, waist circumference; SBP, systolic blood pressure; DBP, diastolic blood pressure; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; PA, physical activity; MVPA, moderate to vigorous physical activity.
[b] Measures based on n = 61.

intensity activities than moderate activity showed no effect on the anthropometry, nor in the blood parameters.

However, regular intake of dairy and nuts showed a negative correlation with BMI while the intake of lean meat showed a positive correlation. Regular intake of dairy was also positively associated with HDL-C and the intake of nuts showed a negative correlation with waist circumference and diastolic blood pressure. Regular intake of eggs was negatively correlated with blood glucose levels (Table 3). The rest of the food groups did not show statistically significant correlations with physical measurements nor with blood parameters.

In the study population, age was negatively correlated with the physical parameters studied. In addition, blood glucose levels were significantly higher in older people. Therefore, we sought to determine how much blood glucose interfered with other variables.

Table 2
Reported dietary habits of women in the study (n = 65) and recommendations of the Sociedad Española de Nutrición Comunitaria (SENC) 2015.

Variable	Mean ± SD	Recommendations
Dietary data		
Potatoes and cereals (servings/day)	3.44 ± 1.47	4–6 servings/day according to physical activity
Vegetables (servings/day)	1.93 ± 0.91	2–3 servings/day
Fruits (servings/day)	2.71 ± 1.30	3–4 servings/day
Olive oil (servings/day)	2.80 ± 0.94	3–6 servings/day
Dairy (servings/day)	2.94 ± 1.09	2–3 servings/day
Fish (servings/week)	3.36 ± 1.48	3–4 servings/week
Lean meat (servings/week)	3.62 ± 1.97	3–4 servings/week
Eggs (servings/week)	2.00 ± 1.22	3–4 servings/week
Legumes (servings/week)	1.93 ± 1.33	3–4 servings/week
Nuts (servings/week)	2.63 ± 2.87	3–7 servings/week
Fat meat (servings/week)	2.63 ± 2.14	Optional, occasional or moderate consumption
Sweets and refined foods (servings/week)	4.37 ± 5.13	Optional, occasional or moderate consumption

3.2. Regression models

3.2.1. Body mass index

Among the regression models built with different parameters of physical activity, the model involving moderate physical activity showed the highest final coefficient of determination ($R^2 = 0.295$; $p = 0.001$). Backward selection was used to determine a model and the first model resulted in high levels of VIF between age and moderate physical activity as they had a strong and negative correlation. In the last model, age was eliminated from the equation and moderate physical activity remained with a significant negative standardized coefficient ($\beta = -0.258$; $p = 0.030$) (Table 4). Regular intake of lean meat was positively associated with BMI ($\beta = 0.289$; $p = 0.019$). However, consuming eggs ($\beta = -0.232$; $p = 0.054$) and nuts ($\beta = -0.211$; $p = 0.076$) was correlated with lower BMI.

3.2.2. Waist circumference

Moderate physical activity was also strongly associated with waist circumference ($\beta = -0.375$; $p = 0.002$) (Table 5). Age was included in the model and high levels of VIF showed its correlation with waist circumference. Backward selection eliminated age in the first step and moderate physical activity remained in a significant model with an adjusted regression coefficient ($R^2 = 0.237$; $p < 0.001$). Standardized coefficients showed that the intake of nuts was related to waist circumference ($\beta = -0.257$; $p = 0.031$). Therefore, we observed that both moderate physical activity and the intake of nuts were associated with smaller waist circumference.

3.2.3. Diastolic blood pressure

Diastolic blood pressure was negatively correlated with moderate physical activity (Table 3). However, linear regression analysis removed it in the first model. As in previous models, the collinearity diagnosis resulted in high levels of VIF. Although a significant model was obtained ($R^2 = 0.159$; $p = 0.043$), the adjusted regression coefficient for each parameter was low and the coefficients of the predictive variables were not significant (Table 6).

3.2.4. HDL-C

Standardized coefficients showed that BMI is the most relevant explanatory parameter in predicting HDL-C levels ($\beta = -0.364$; $p = 0.003$) (Table 7). As in the previous model, the collinearity diagnosis resulted in high levels of VIF. After applying backward selection, the variable age was eliminated from the equation and the number of steps per day remained in the model. Although the intake of eggs was associated with high levels of HDL-C (Table 3) and the correlation between HDL-C and age was nearly significant (Table 3), these variables were not included in the last equation ($R^2 = 0.331$; $p < 0.001$). Further, the steps per day ($\beta = 0.221$; $p = 0.059$) and the intake of dairy ($\beta = 0.197$; $p = 0.094$) were also close to being significant (Table 7).

3.2.5. Blood glucose

A logistic regression model was performed with parameters significantly correlated with blood glucose (Table 8). A high regression coefficient was obtained in the equation ($R^2 = 0.523$; $p = 0.001$). Time spent in moderate physical activity, egg intake, and age predicted blood glucose levels. However, BMI was excluded, as its coefficient was not significant.

4. Discussion

In a cohort of 65 older women, we observed that the combination of physical activity and certain dietary patterns were good predictors of cardiovascular risk factors including BMI, waist circumference, blood glucose, HDL-C levels, and diastolic blood pressure. We obtained statistically significant models that explain the relationship between physical activity and the previously mentioned parameters. In addition, we

Table 3
(A) Pearson's correlation coefficients (r) between body-mass index, waist circumference, diastolic blood pressure, high-density lipoprotein, age, and anthropometric, physical, and dietary parameters. (B) Spearman's correlation coefficients (rho) between glucose, age, and anthropometric, physical and dietary parameters.

Variable[a]	Age (years)	BMI (kg/m^2)	cpm	spd	Moderate PA (min/day)	Dairy (servings/day)	Lean meat (servings/week)	Eggs (servings/week)	Nuts (servings/week)
(A)									
BMI (kg/m^2)	r = 0.072 $p = 0.567$	r = 1	**r = −0.323*** **$p = 0.011$**	**r = −0.272*** **$p = 0.034$**	**r = −0.344**** **$p = 0.007$**	**r = −0.289*** **$p = 0.019$**	**r = 0.336**** **$p = 0.006$**	r = −0.139 $p = 0.270$	**r = −0.308**** **$p = 0.009$**
WC (cm)	r = 0.219 $p = 0.079$	**r = 0.834***** **$p < 0.000$**	**r = −0.408***** **$p = 0.001$**	**r = −0.387***** **$p = 0.002$**	**r = −0.413***** **$p = 0.001$**	r = −0.238 $p = 0.056$	r = 0.214 $p = 0.086$	r = −0.182 $p = 0.148$	**r = −0.339**** **$p = 0.006$**
DBP (mm Hg)	r = 0.119 $p = 0.345$	**r = 0.347***** **$p = 0.004$**	r = −0.197 $p = 0.128$	r = −0.195 $p = 0.132$	**r = −0.260*** **$p = 0.043$**	r = −0.091 $p = 0.470$	r = 0.061 $p = 0.629$	r = −0.189 $p = 0.132$	**r = −0.327*** **$p = 0.016$**
HDL-C (mg/dL)	r = −0.212 $p = 0.093$	**r = −0.436****** **$p < 0.000$**	**r = 0.338**** **$p = 0.008$**	**r = 0.334**** **$p = 0.009$**	**r = 0.354**** **$p = 0.006$**	**r = 0.341**** **$p = 0.006$**	r = −0.126 $p = 0.321$	r = 0.246 $p = 0.050$	r = 0.061 $p = 0.400$
(B)									
Glucose (mg/dL)	**rho = 0.258*** **$p = 0.038$**	rho = 0.220 $p = 0.077$	**rho = −0.351**** **$p = 0.006$**	**rho = −0.276*** **$p = 0.031$**	**rho = −0.346**** **$p = 0.006$**	rho = 0.140 $p = 0.267$	rho = 0.017 $p = 0.892$	**rho = −0.331**** **$p = 0.007$**	rho = −0.159 $p = 0.207$

Significant results in bold.
[a] BMI, body mass index; cpm, counts per minute; spd, steps per day; PA, physical activity; WC, waist circumference; DBP, diastolic blood pressure; HDL-C, high-density lipoprotein cholesterol.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.005$.
**** $p < 0.001$.

observed that in our sample, physical activity and diet were better predictors of cardiovascular risk than age.

Moderate intensity physical activity is one of the best predictors of cardiovascular risk, such as BMI, waist circumference, diastolic blood pressure, and fasting plasma glucose. Consistent with our results, other authors have observed that an increase in physical activity intensity is associated with improvement in several CVD risk factors, in both postmenopausal women (Manson et al., 2002) and obese young women (Racil et al., 2013). However, the methods used to quantify physical activity and the variables considered in these studies were not exactly the same as we describe here. In most of the previous studies, physical activity was measured using questionnaires while our study utilized accelerometers. In a study conducted in young and middle-aged adults, Andersson et al. (2015) observed that MVPA, measured by accelerometry, was associated with beneficial effects on the cardiovascular system (Andersson et al., 2015). In addition, Van der Velde et al. (2015), in a cross-sectional study in adults aged 18–49 years, suggested that a combination of increasing MVPA and fitness was more beneficial to decreasing cardio-metabolic risk than decreasing sedentary time (Van der Velde et al., 2015). A meta-analysis of prospective cohort studies concluded that "high levels of leisure time physical activity and moderate levels of occupational physical activity had a beneficial effect on cardiovascular health by reducing the overall risk of incident coronary heart disease and stroke among men and women" (Li and Siegrist, 2012). Recently, The Cardiovascular Health Study concluded that greater physical activity and exercise intensity were associated with lower risks of coronary heart disease, stroke, and CVD in older adults (Soares-Miranda et al., 2015). Our results also suggest that in older women, moderate intensity of physical activity is associated with favorable changes in risk factors for CVD and blood glucose levels compared to lower intensities.

The linear regression model we created using steps per day explained the blood levels of HDL-c better than the model we created using physical activity intensities. Other studies carried out in both older adults (Stewart et al., 2005) and older women (Carvalho et al., 2010) also reported that exercise training was associated with increased HDL-C, although they did not quantify objectively the nature of the physical activity. On the other hand, some studies observed that in older adults a walking-based program increased the levels of HDL-C (Fahlman et al., 2002). Other studies, though, reported no significant change (Cottell et al., 2011; Marques et al., 2009) and some reported a decrease in HDL-C after the resistance training (Arnarson et al., 2014; Williams et al., 2011). This could indicate that in older people, the amount of physical activity performed could have a higher influence on HDL-C than the intensity or the resistance of the activity.

We also observed that nutritional predictive variables were nuts, dairy, and eggs. The consumption of these protein-rich food groups predicts favorable changes in risk factors for CVD and blood glucose levels in our sample. It is important to note that these foods were considered not to be healthy until recently because they are quite rich in fat. Similar

Table 4
Linear regression model built with the dependent variable body mass index including moderate physical activity (PA) and significantly correlated food groups as independent variables.

	B	SE	Standard coefficient (β)	*t*	*p* value
(Constant)	29.004	1.370		21.169	0.000
Moderate PA	−0.024	0.011	−0.258	−2.231	0.030
Lean meat	3.609	1.493	0.289	2.416	0.019
Eggs	−4.750	2.415	−0.232	−1.967	0.054
Nuts	−1.833	1.015	−0.211	−1.806	0.076

ANOVA. $R^2 = 0.295$; $p = 0.001$.
B, Non-standardized coefficient; SE, standard error; *t*, *t*-test statistics; *p* value indicates the significance of an ANOVA checking the value of the predictive model; R^2, coefficient of determination.

Table 5
Linear regression model built with the dependent variable waist circumference including moderate physical activity (PA) and significantly correlated food groups as independent variables.

	B	SE	Standard coefficient (β)	*t*	*p* value
(Constant)	96.139	2.372		40.526	0.000
Moderate PA	−0.092	0.029	−0.375	−3.234	0.002
Nuts	−5.795	2.62	−0.257	−2.212	0.031

ANOVA. $R^2 = 0.237$; $p < 0.001$.
B, Non-standardized coefficient; SE, standard error; *t*, *t*-test statistics; *p* value indicates the significance of an ANOVA checking the value of the predictive model; R^2, coefficient of determination.

Table 6
Linear regression model built with the dependent variable diastolic blood pressure including moderate physical activity (PA) and significantly correlated food groups, age, and body mass index (BMI) as independent variables.

	B	SE	Standard coefficient (β)	*t*	*p* value
(Constant)	55.120	18.362		3.002	0.004
Moderate PA	−0.026	0.036	−0.109	−0.733	0.467
Nuts	−3.737	2.905	−0.169	−1.286	0.204
Age	0.147	0.194	0.107	0.756	0.453
BMI	0.563	0.347	0.222	1.621	0.111

ANOVA. $R^2 = 0.159$; $p = 0.043$.
B, Non-standardized coefficient B; SE, standard error; *t*, *t*-test statistics; *p* value indicates the significance of an ANOVA checking the value of the predictive model; R^2, coefficient of determination.

results were published by the "PREvención con DIeta MEDiterránea" (PREDIMED) trial, which concluded that in older people, a Mediterranean diet supplemented with extra virgin olive oil or nuts reduced 24-hour ambulatory blood pressure, total cholesterol, and fasting glucose compared to a control, low-fat diet (Domenech et al., 2014). Similar to our study, nuts have shown a protective effect on cardiovascular risk and hypertension (Casas-Agustench et al., 2011; Martinez-Lapiscina et al., 2010). In a study led by Brown et al. (2015), nut consumers had significantly lower weight, BMI, and waist circumference than non-nut consumers (Brown et al., 2015). Our study confirms the results from larger studies reporting that nut consumption is associated with better outcomes for a number of cardiovascular risk factors. Dairy products, however, remain controversial. There are some studies showing the beneficial role of dairy products in protecting against cardiovascular disease by lowering values of glycemia and hypertension (Da Silva et al., 2014; Louie et al., 2013). However, there are a few studies where results were not statistically significant, leaving doubts about the protective role of dairy consumption (Bendtsen et al., 2014; Fard et al., 2015). Regarding eggs, our results are consistent with other recently published studies. The study led by Lee et al. (2013) concluded that egg consumption was inversely associated with insulin sensitivity (Lee et al., 2013), while a study led by Woo et al. (2016) showed that higher egg consumption was correlated with lower blood glucose levels (Woo et al., 2016).

We observed that physical activity, as well as the ability to perform some types and intensities of physical activity decreases as the age of the participants progresses, which is widely supported by other studies (Lohne Seiler et al., 2014; Shiroma et al., 2013). However, in our study we found that the dietary pattern held by this population of women was not correlated with the age. This corroborates what others have found, suggesting that eating habits are set with age and seniors are reluctant to change their dietary habits (Fernyhough et al., 1999; Marrodán Serrano et al., 2003). Therefore, one of the objectives could be promoting healthy eating habits as recommended in the Mediterranean diet, because these habits could lower cardiovascular risk even in older people whose physical activity is low.

Table 7
Linear regression model built with the dependent variable high-density lipoprotein cholesterol including body mass index (BMI), steps per day, and significantly correlated food groups as independent variables.

	B	SE	Standard coefficient (β)	*t*	*p* value
(Constant)	94.347	18.528		5.092	0.000
BMI	−1.649	0.541	−0.364	−3.050	0.003
Steps per day	0.001	0.000	0.221	1.925	0.059
Dairy	2.974	1.743	0.197	1.706	0.094

ANOVA. $R^2 = 0.331$; $p < 0.001$.
B, Non-standardized coefficient B; SE, standard error; *t*, *t*-test statistics; *p* value indicates the significance of an ANOVA checking the value of the predictive model; R^2, coefficient of determination.

Table 8
Logistic regression model built with the dependent variable blood glucose including moderate physical activity (PA) and significantly correlated food groups and age as independent variables.

	B	SE	Wald	df	*p* value
(Constant)	22.2855	9.866	5.102	1.000	0.024
Moderate PA	−0.0844	0.038	4.983	1.000	0.026
Eggs	−0.2885	4.907	3.584	1.000	0.058
Age	−0.2572	0.121	4.539	1.000	0.033

Chi-Square $R^2 = 0.523$; $p = 0.001$.
B, Non-standardized coefficient B; SE, standard error; df, degrees of freedom; *p* value resulted from a χ^2 test checking the value of the predictive model; R^2, Nagelkerke's coefficient.

5. Conclusions

The combination of moderate physical activity and a diet including protein-rich foods as nuts, dairy, and eggs were better predictors associated with favorable changes in risk factors for CVD than age in our sample.

Taking into account that physical activity decreases over the lifespan, our findings also indicate that maintaining a healthy diet was of vital importance at this stage of life because, regardless of age, dietary habits were associated with better cardiovascular risk parameters.

Further research is needed to expand the knowledge and to strengthen the results found in this study.

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Conflict of interest

All authors declare that they have no conflict of interest.

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